

Operating Systems – EDA093/DIT401

Course Introduction

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UNIVERSITY OF
GOTHENBURG

Agenda

- Course homepage & support team
- Main textbook
- Why study Operating Systems?
- What is this course about?
 - What is this course NOT about?
- Course organization
- Best practices
- Student representatives

Course homepage & support team

<https://chalmers.instructure.com/courses/35797>

Examiners / Lecturers:

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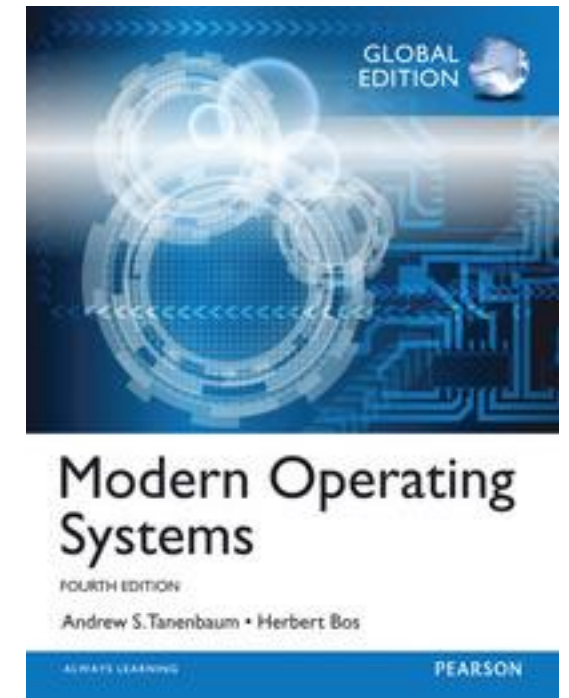
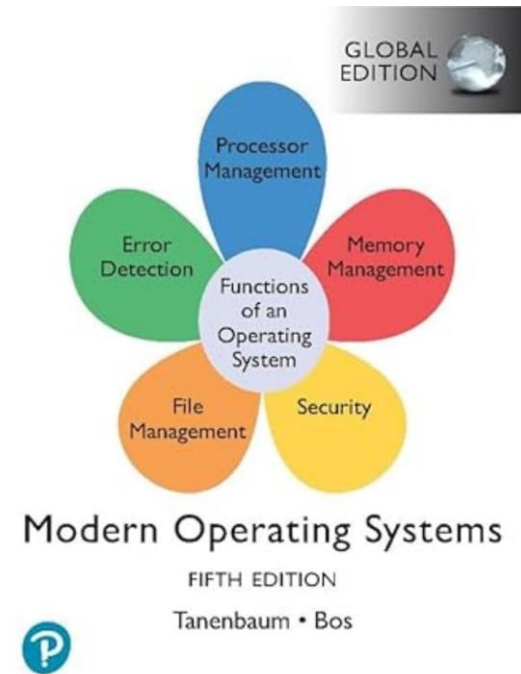
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Main textbook

Modern Operating Systems by Andrew S. Tanenbaum
(fifth edition)

There's a fourth edition too,
both are OK



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ROBOCOP



```

COMMAND.COM
LOAD BIOS
BIOS SYSTEM CHECK
RAM CHECK
CONFIG.SYS
BIO.COM INTERFACE
TO ROM I/O
CONTROLLER
COMSPEC.EXE
MEMORY.DAT
ROBO UTILS
SYSTEM BUFFER
PARAMETERS
PARITY SET
MEMORY SET
SYSTEM STATUS
OK_
    
```

```

#85 A8 8A
#87 B9 E1 60
#8A 99 F8 83
#8D 8C
#8E 18 F7
#90 AD F3 83
#93 49 A5
#95 8D F4 83
    
```

```

#98 A9 83
#9A A8 F8
    
```

```

29 SLOTA EQU $06
30 SLOT EQU $07
31 BANK EQU $08
32 COUNT EQU $F9
33 VIEW EQU $FA
34 VIEWM EQU $FB
35 STOR EQU $FC
36 STORM EQU $FD
37 D8 EQU $FE
38 D8H EQU $FF
    
```

```

ADDRESS ADDR (
<<< POKE RAMCARD
<<< POKE 16K RAM
HIBYTE # OF POIN
PAGE VIEWED (IM
HIBYTE OF VIEW
PAGE IN STORAGE
HIBYTE OF STOR
USED FOR BANK S
HIBYTE OF D8-B
    
```

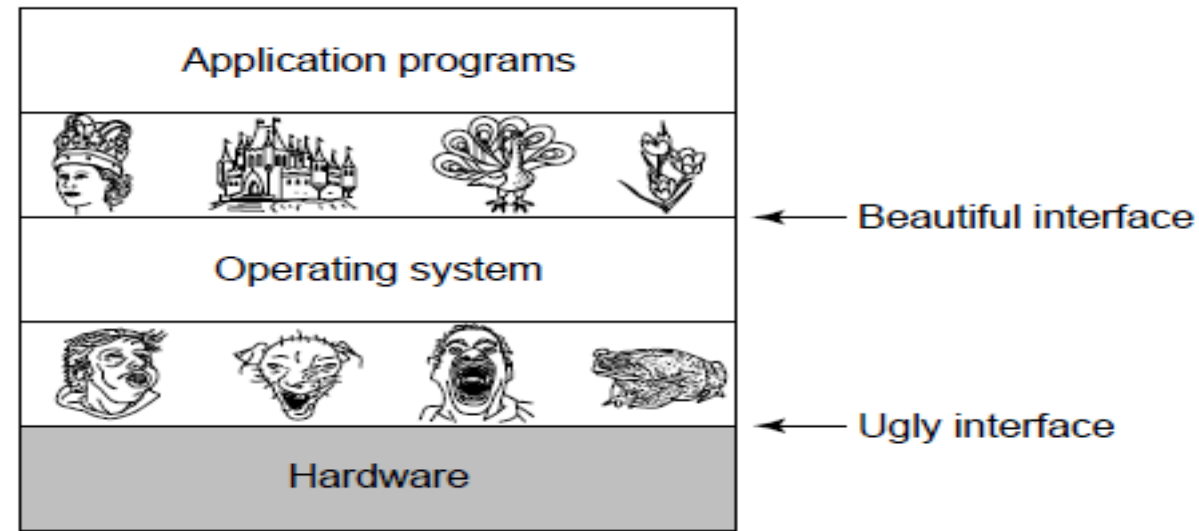
Why study Operating Systems?

Where there are computers, there are operating systems...



Why study Operating Systems?

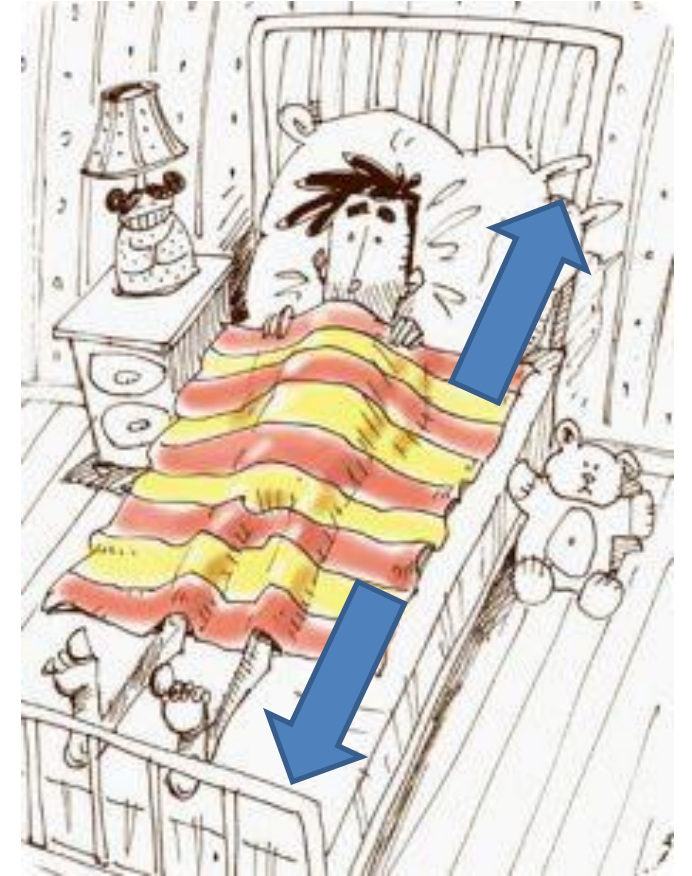
- Provides services to system users
- “Shields” the user from the hardware
- Resource manager:
 - CPU(s)
 - memory
 - I/O devices
 -



Picture source: Modern OS, by A. Tanenbaum

Why study Operating Systems?

- To learn about system functionality and design
 - performance vs. simplicity, HW vs. SW, etc.
- To learn how to manage complexity through appropriate abstractions
- again: Because computers and OSs are everywhere!



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What is this course not about?

What is this course not about?

- Learn how to use Windows
- Learn how to use Linux
- Learn how to use OS X
- Learn how to program in C



What is this course about, then?

Content of the course:

- Introduction / System Structures
- Processes / Threads
- Multithreaded Programming
- Process scheduling
- Synchronization
- Deadlocks
- Memory Management
- Virtual Memory
- File Systems
- I/O Systems
- Security / Protection
- Virtualization

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Course organization

- Lectures, Tutorials, Exercises sessions
 - “Lectures and Schedule” @ web page
 - Programming assignments
 - “Lab1, Lab2, Lab3” @ web page; **(more about this in the following)**
- Exam
 - 28th October 2025, 5th January 2026, 17th August 2026.

Let's check the Canvas page...

About the labs

Let's check the Canvas page again...

About the labs

- There will be 3 labs. For each lab:
 - The material is made available in Canvas and presented in dedicated pages. For each lab, each group needs to pass a preparation test.
 - Groups must complete each lab using the `distans.cdal.chalmers.se:33899` servers (links will be given). Groups can practice of course with other computers, but the TAs' final evaluation is carried out only for code that is compiled / run at those servers
 - a script will be provided for each lab to check (@ `distans.cdal.chalmers.se`) whether the lab compiles / runs / passes a certain set of tests
 - Successfully running the script does not imply the group has passed the lab.
 - There can be manual inspection and groups need to explain some of the decisions they took and/or modify their code to pass the lab. Passing the script is only an indication that the group can be further inspected by a TA. At the presentation, we decide who explains

About the labs

... There will be 3 labs. For each lab:

- Once the script runs successfully, groups are expected to upload to Canvas their source files, the message generated by the script **and a report**.
- Each lab has an internal deadline (2 weeks after the lab introduction, approximately).
- Each group is strongly recommended to book at least one slot during the lab sessions, slots are available in Canvas.
- Each group books its slot using the group id
- You can ask stuff to the TAs even if you do not book, but only if they are not booked when you want to ask, priority is given to booked ones
- If you have questions about previous labs, do not use booking, only enter and wait for a TA (priority is always given to the current lab)
- If a group fails the lab, the resubmission will be at the next re-exam (for that specific lab)

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Best Practices

1. Participate to lectures, exercise sessions and labs
 1. Ask when something is not clear
 2. Ask when you have some doubt about what is being discussed
2. Study at pace with classes, do the labs in time
3. Use the book and other material
 1. Read the book
 2. Try the exercises in the book
4. Be nice to future generations
 1. Provide feedback
 2. Evaluate the course!!!
5. Check the updates in the Canvas page
6. Look at messages in Canvas
7. **PLAY WITH UNIX – PLAY WITH GIT – PLAY WITH GITHUB**

Best Practices

- Do you know what this is?
 - ls
 - mv *filename1 filename2*
 - chmod *options filename*
 - man *commandname*
 - ls -l | grep "Aug" | sort -n
- Install a virtual machine (e.g., Virtual Box) → Install Unix OS (e.g., Ubuntu) → start playing with commands and system calls

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Student representatives

- Student representatives:
 - Pengfei Li pengfeil@student.chalmers.se
 - Shuo Zhang shuozha@student.chalmers.se
 - Fredrik Österström freoster@student.chalmers.se
 - Reza Amani rezaam@student.chalmers.se

Student representatives

TASK: responsible for discussing about the course together with the examiner/course responsible and the program board of the program that gives the course, as follows:

1. The student representatives and the course responsible have **two informal meetings during the course**: one in the second study week (get acquainted) and one after approximately half the course (give feedback to the teacher on the progress of the course).
2. After the course, an evaluation questionnaire is sent out. The examiner will get an opportunity to add extra questions to the standard questionnaire.
3. In the middle of the next study period, there will be an **evaluation meeting**: the student representatives and the examiner/course responsible meet the board of the program who gives the course, discussing the course's advantages and potential for improvement.

The student representatives participating in the final meeting get a gift certificate valid at Cremona.

Academic integrity and honesty



Hans Hjelmgren, Head of program EE,
Associate Prof at MC2

Studies

[Incoming student 2016](#)[Tuition Fees and registration for fee paying students](#)[The academic year - dates and times](#)[Course Information](#)[Degree Certificate](#)[Examinations](#)[Incoming exchange students](#)[Joint rules and directives](#)[Master's Thesis](#)[Programme information](#)[Student work environment](#)[Study in an exchange programme](#)[Study results and certificates](#)

Joint rules and directives for the undergraduate programmes

As a student at Chalmers you are covered by rights and rules related to your studies.

Chalmers is a foundation since 1994 and university activities are run in the form of a wholly-owned limited company, Chalmers tekniska högskola AB. Consequently, Chalmers does not come under the Higher Education Ordinance.

The supreme decision-making body at Chalmers is its board, which on behalf of the University reaches final decisions in dealings between the student and the University.

For the undergraduate programmes, Chalmers has adopted Rules of Procedure and a Disciplinary Code. The President has also decided on guidelines, rules and other regulations for undergraduate programmes.

Swedish law has the principle of public access to official records. Briefly, this means that the public and the media have transparency in Chalmers' activities and have the right to inspect public documents.

Policy documents at Chalmers

- [Academic integrity and honesty](#)
- [Admission Regulations for Bachelor's \(undergraduate\) and Master's level \(graduate/advanced\)](#)
- [Complaints](#)
- [Directions for the use of the IT-resources](#)
- [Guidelines for applying for recognition of entire courses](#)
- [Openness and insight](#)
- [Local regulations regarding scholarships](#)
- [Request for a change of examiner](#)
- [Regulations for teaching and research faculty](#)
- [Rules of Discipline](#)
- [Rules for examinations](#)
- [Rules for Master's Thesis](#)
- [Rules for payment of tuition fees](#)
- [Rules of procedure for undergraduate programmes](#)
- [Scholarship regulations](#)

[CONTACT PAGE RESPONSIBLE](#)

Published: Mon 21 Sep 2009. Modified: Mon 25 Apr 2016

Academic Honesty and Integrity at Chalmers –

What Are the Rules of the Game?

Academic integrity and honesty

- The document in the previous slide will give you examples of how to use and refer to the knowledge of others.
- We expect you to read the document on *Academic integrity and honesty* and hand it out today!
- In addition to this, we will give you some examples from the document in the following slides

Academic integrity and honesty

To study is to turn the knowledge of others into one's own knowledge.

- An important motivating force behind academic work is “getting the credit” for one's advances
- There are rules for how recognition is paid to the contributions of colleagues and predecessors
- Cheating can be collaborating on a certain assignment beyond the rules of the game, omitting trial experiments that do not “fit in”, using aids for an examination assignment that have not been approved, plagiarism

Definition of Plagiarism

“Claiming someone else’s work as your own work”

J. Carroll and C.-M. Zetterling,
Guiding students away from plagiarism, KTH, 2009.

Examples of plagiarism

- Hand in someone else's work as your own
- Copy words or ideas from someone else without referring to the source
- Not use quotation marks for words that are cited.
- Excessive borrowing of the structure of a text may also result in plagiarism. Even if you cite your source, do not produce a text that is identical to the original text.

Important

- As the writer you are responsible that your papers do not plagiarize the works of others
- It is customary to be able to discuss an assignment with one's classmates; however, every student must submit his or her answers individually
- For larger project assignments, it is customary that students divide the tasks and then assemble their separately written sections into one larger team report. However, every author is responsible for the *entire* report, i.e. the mistake of one of the authors affects everyone on the team.

You are here to learn!

- You don't need to know everything from the start
- Always ask your teacher if you are not certain
- You must be able to discuss your *draft* reports and hand-ins without running the risk of being accused of plagiarism and cheating.

Rules of Discipline

Rules of Discipline

- The purpose of the Rules of Discipline is to promote equality between students, good conduct, a good study and social environment and adherence to the rules and directives at Chalmers.
- The Rules of Discipline stipulate the type of conduct and actions that can lead to disciplinary action. In addition, they stipulate the grounds on which disciplinary action may be taken at Chalmers and the procedure for such action.
- You can find the Rules of Discipline and other rules and directives in the Student Portal

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The Disciplinary Committee

The Disciplinary Committee makes decisions regarding disciplinary action.

The Disciplinary Committee at Chalmers consists of:

- two members learned in the law, at least one of whom is a judge
- the Vice-President for undergraduate and master programs
- two faculty members
- two students

Disciplinary action

A student may be **warned** or **suspended** if he or she

- attempts to mislead in connection with an examination (=cheating)
- disturbs teaching or an examination
- insults the integrity of other student or employee at Chalmers
- otherwise breaches the Chalmers rules

The Disciplinary Committee can decide on **expulsion** in very serious case for example if the student is guilty of serious criminal activity

The Disciplinary Committee and its work

- Students, faculty members and other employees are jointly responsible for reporting the matter to the President without delay if there are grounds for suspecting an infringement or other misconduct.
- Once the report has been received by the President, the person to whom the report pertains shall be informed promptly in writing at his or her last known address.
- The student shall be given a copy of the report and have an opportunity to present his or her account of the matter. The student ombudsman shall be provided with a copy of the report and the material from the investigation.

The Disciplinary Committee and its work

- If the student wishes to present a verbal account of the case to the Disciplinary Committee, he or she shall be able to do so.
- The student shall also have the opportunity to be represented by a person whom he or she trusts. The student ombudsman should contact the student to inform him or her about the procedure and the assistance available to the student.
- The Disciplinary Committee should give reasons for any decision.
- The student may appeal to the Chalmers Board against a Disciplinary Committee decision regarding a warning, suspension or expulsion.

Thank you for your attention!

Questions?